

Socioeconomic Inequalities And Determinants Of Modern Contraceptive Use Among Ever-Married Women Of Reproductive Age In Bangladesh



Course Code: STA 4207

**A research project submitted as a partial fulfillment of the requirements for
the degree of Bachelor of Science in Statistics**

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LETTER OF TRANSMITTAL

Date: 20 August 2026

To

Prof. Dr. Muhammad Tareq
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Dhaka-1100, Bangladesh

Subject: Submission of Undergraduate Research Project Report

Dear Sir,

With due respect, we, the undergraduate students of the Department of Statistics, Jagannath University, are pleased to submit our research project report titled: **“Socioeconomic Inequalities and Determinants of Modern Contraceptive Use Among Ever-Married Women of Reproductive Age in Bangladesh.”**

This research project report has been prepared as a partial fulfillment of the requirements for the course Research Project (Course Code: STA 4207) under the B.Sc. (Honors) in Statistics program for the session 2020–21. The project was carried out under the supervision of **Prof. Dr. Abu Sayeed Md. Ripon Rouf**, Professor, Department of Statistics. We would like to express our sincere gratitude to our honorable supervisor for his guidance, advice, and continuous support throughout the project.

We will be highly obliged if you kindly receive this report and provide your valuable evaluation.

Sincerely,

MD. Habibur Rahman
ID: B200304038

APPROVAL

This is to certify that the project report titled: “**Socioeconomic Inequalities and Determinants of Modern Contraceptive Use Among Ever-Married Women of Reproductive Age in Bangladesh.**” submitted by MD. Habibur Rahman (ID: B200304038) has been examined and approved as a satisfactory fulfillment of the requirements for the course Research Project (STA 4207) in the Department of Statistics, Jagannath University.

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DECLARATION

This is to declare that this project is our original work. No part of this work has been submitted, either partially or fully, for the award of any other degree or diploma. All project-related information will remain confidential and shall not be disclosed without the formal consent of the project supervisor. Relevant previous works cited in this report have been properly acknowledged. We have adhered to the plagiarism policy as stated by the supervisor.

Students' names & Signatures

.....

MD. Habibur Rahman

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Finally, we express our heartfelt thanks to our parents, family members, friends, and everyone who directly or indirectly supported, encouraged, and inspired us throughout this research journey. Their constant motivation and goodwill have been invaluable, and we sincerely appreciate their contributions to the successful completion of this study.

ABSTRACT

Population control and effective family planning remain vital components of maternal and child health policy in Bangladesh. Although overall contraceptive awareness has grown over recent decades, reliance on traditional methods persists, and significant regional and socioeconomic disparities hinder the widespread adoption of modern contraceptives. Utilizing individual-level nationally representative data from the Bangladesh Demographic and Health Survey (BDHS) 2022 dataset ($n = 11,452$), a data-driven quantitative analysis was conducted. Current contraceptive use served as the primary dependent variable, evaluated against key demographic, social, and economic factors including maternal age, place of residence, household wealth index, and mass media exposure. Statistical techniques involved univariate percentage distributions, bivariate Chi-square (χ^2) tests, multivariate Binary Logistic Regression, and socioeconomic inequality metrics (Concentration Index and Wagstaff Index). The overall modern contraceptive prevalence rate was found to be 72.16%. Binary logistic regression confirmed that urban residency, higher household wealth, and mass media exposure (especially internet access) are significantly associated ($p < 0.05$) with higher odds of adopting modern contraceptive methods. Women using the internet demonstrated up to 62% higher odds of adopting modern methods compared to non-users. Inequality analysis via the Wagstaff Corrected Index (WCIX = +0.079, $p < 0.001$) confirmed persistent pro-rich inequality in modern contraceptive uptake. This study underscores the necessity of targeted public health interventions, digital health campaigns, expanded LARC subsidies for poorer quintiles, and strengthened rural healthcare infrastructure to accelerate modern contraceptive adoption in Bangladesh.

Keywords: Modern contraceptives, family planning, BDHS 2022, logistic regression, ever-married women, Bangladesh, public health.

CHAPTER ONE
INTRODUCTION

1.1 Background of the Study

Rapid population growth poses significant challenges to socioeconomic development, resource distribution, and public health infrastructure in developing nations. In Bangladesh, managing population growth has long been a national public health priority. Family planning programs are acknowledged not only as a mechanism for population control, but also as a fundamental strategy for improving maternal and child health, reducing maternal mortality, and empowering women.

Over the past few decades, Bangladesh has made commendable progress in expanding access to family planning services. The national Contraceptive Prevalence Rate (CPR) has increased steadily, contributing to a noticeable decline in the Total Fertility Rate (TFR). However, despite these overarching national successes, significant gaps remain between overall contraceptive knowledge and the sustained practice of effective family planning.

A critical issue facing the country's public health landscape is the continued reliance on traditional, less reliable contraceptive methods alongside persistent non-use in various sub-populations. Modern methods offer significantly higher efficacy in preventing unintended pregnancies and their associated health risks compared to traditional methods (e.g., withdrawal or rhythm methods).

The uptake of modern contraceptives is deeply intertwined with complex socio-demographic and economic determinants:

- **Educational Attainment:** Literacy and formal education empower women with accurate reproductive health information, enhance autonomy in decision-making, and promote positive attitudes toward modern family planning.
- **Geographic Disparities:** Rural and urban disparities persist in healthcare delivery. Urban women often benefit from greater proximity to healthcare facilities, private providers, and pharmacy networks, whereas rural women frequently face service delivery bottlenecks and geographic barriers.
- **Socioeconomic Status:** Household wealth influences a family's ability to afford contraceptive supplies, travel to clinical centers, or access higher-quality private care options.
- **Mass Media Exposure:** Mass media platforms—including television, radio, and digital channels—play a pivotal role in dispelling cultural myths, normalizing reproductive health discussions, and improving health literacy across diverse demographics.

1.2 Rational of the Study

Despite significant national strides in improving reproductive health indicators, Bangladesh faces a pivotal phase in its demographic transition. While awareness of family planning is nearly universal across the country, a substantial gap persists between knowledge and the actual, consistent adoption of effective modern contraceptives. This gap leads to unintended pregnancies, shorter birth intervals, unsafe abortions, and elevated risks of maternal and child mortality, presenting a critical challenge to the country's public health system.

A primary issue driving this research is the ongoing reliance on traditional family planning methods, such as withdrawal or periodic abstinence, despite the widespread availability of modern alternatives like pills, injectables, implants, IUDs, and condoms. Because traditional methods carry significantly higher failure rates, women who rely on them face elevated health and financial risks. Unpacking the underlying reasons why women revert to or prefer traditional methods over modern alternatives is essential for updating national family planning strategies.

Furthermore, much of the existing literature relies on older survey cycles, creating a need for updated evidence that reflects recent post-pandemic demographic realities. Utilizing the BDHS 2022 dataset offers a crucial opportunity to evaluate contemporary trends in contraceptive choices, socioeconomic dynamics, and healthcare access among ever-married women aged 15–49. Up-to-date data is mandatory for accurate, responsive policymaking.

This investigation is further necessitated by persistent socioeconomic and regional inequities that are often masked by broad national averages. Rural-urban divides, varying educational attainment levels, household wealth gaps, and unequal exposure to mass media heavily influence a woman's reproductive autonomy. By applying rigorous statistical modeling—such as Chi-square tests and Binary Logistic Regression—this study identifies which specific subgroups remain underserved, enabling health authorities to allocate limited public resources efficiently.

Ultimately, this study fills a crucial knowledge gap by pinpointing the precise sociodemographic barriers to modern contraceptive use in contemporary Bangladesh. The empirical evidence generated provides direct support for national health sector operational plans and aligns with Sustainable Development Goal 3.7, laying the groundwork for more responsive, equitable, and effective reproductive health policies.

1.3 Objectives of the Study

General Objective

To examine the pattern and determinants of modern contraceptive use (MCU) among currently married women in Bangladesh using BDHS data

Specific Objectives:

1. To identify the prevalence and method distribution of modern contraceptive use.
2. To analyze the effects of age, place of residence (rural vs. urban), household wealth, and media exposure on contraceptive use.
3. To quantify the extent of socioeconomic inequalities in modern contraceptive adoption using concentration indices.

1.4 Research Questions

1. What is the current proportion of modern contraceptive use among ever-married women in Bangladesh?
2. What are the background characteristics that significantly affect the adoption of modern contraceptive methods in Bangladesh?
3. Is there any relationship between modern contraceptive use and a respondent's place of residence, wealth status, and exposure to mass media/internet?

1.5 Research Hypotheses

- H_0 : There is no significant relationship between a respondent's place of residence (urban vs. rural) and their use of modern contraceptive methods.
- H_0 : There is no significant relationship between mass media/internet exposure and the adoption of modern contraceptive methods.
- H_0 : There is no significant relationship between household wealth quintiles and modern contraceptive adoption.

1.6 Limitations of the Study

However, it is essential to acknowledge certain limitations. Firstly, due to data constraints in the secondary dataset, the study couldn't incorporate several significant factors pivotal in determining modern contraceptive use, such as detailed husband/partner attitudes, specific biological side effects experienced, and health facility supply dynamics. Secondly, given the cross-sectional design of the Bangladesh Demographic and Health Survey, establishing direct cause-and-effect relationships between sociodemographic predictors and contraceptive adoption was not feasible. These constraints should be considered when interpreting the findings, and future research may benefit from longitudinal designs to further enrich our understanding of the dynamics involved in modern contraceptive choices in Bangladesh.

CHAPTER TWO
REVIEW OF LITERATURE

2.1 Introduction

The current chapter serves as a comprehensive review of both published and unpublished documents relevant to the study's subject. The main objective is to examine the insights offered by other researchers on the key concepts related to the subject under investigation. Specifically, the focus is on studying the association between various levels of factors and the adoption of modern contraceptive methods (MCM) in the context of Bangladesh.

2.2 Review of Literature

The study by **Kundu et al. (2022)** conducted in Bangladesh revealed a significant association between sociodemographic predictors and the choice of family planning methods among women of reproductive age. Notably, the analysis demonstrated that place of residence played a crucial role, with rural women being 21% less likely to utilize modern contraceptive methods compared to their urban counterparts. The research emphasized the mediating role of male partner involvement, indicating that husband decision-making positively influenced the adoption of modern contraceptives.

In a study focusing on modern contraceptive practices in Bangladesh, **Rahman et al. (2025)** discovered that socioeconomic status heavily influenced modern contraceptive adoption. Their analysis identified several predictors influencing modern contraceptive utilization, including mother's education, husband's education, wealth status, parity, media exposure, place of residence, division, and occupational characteristics. Moreover, educational attainment was highlighted as a dominant predictor, with highly educated women being 1.64 times more likely to adopt modern contraceptive methods compared to those with no formal education.

Pratley's comprehensive review (2016) underscored the strong association between women's empowerment and maternal healthcare utilization, including family planning choices. The study highlighted existing gaps in knowledge related to the aggregation of empowerment indicators, the absence of health decision-making dimensions, and the need for indicators measuring psychological autonomy. The findings emphasized the importance of indicators beyond individual-level measurements of empowerment in evaluating reproductive health behaviors.

Tekelab et al. (2019), in their study on reproductive healthcare utilization, demonstrated that empowering women, increasing accessibility to family planning outlets, enhancing partner participation, media exposure, and advocating family planning through mass media all contributed to increased adoption of modern contraceptive methods. Additionally, the study found that women with higher autonomy and health literacy were far more likely to abandon traditional methods in favor of highly effective modern alternatives.

A study in Bangladesh by **Kibria et al. (2023)** highlighted persistent regional disparities in the uptake of modern contraceptive methods among late reproductive-aged women. The influential variables included woman's education level, husband's education level, wealth status, geographic division, and regular mass media access. The results indicated that economic conditions, measured by the household wealth index, alongside regular exposure to electronic media, played a crucial role in determining compliance with modern family planning practices.

Islam et al. (2023) study in Bangladesh revealed that while short-acting reversible contraceptives (SARC), such as oral pills, dominated the landscape of modern method users, long-acting reversible contraceptives (LARC) and permanent methods remained significantly underutilized. Modern contraceptive choice was found to be strongly conditioned by maternal age, parity, and formal education, with less-educated and higher-parity women displaying different method preferences compared to younger, urban women.

Chanda et al. (2020) study in Bangladesh highlighted the significance of media exposure, wealth index, and employment of both mothers and husbands as factors influencing the adoption of healthcare and family planning services. Women with regular exposure to mass media (such as television and radio) were substantially more likely to adopt modern contraceptive methods, and economic status emerged as a critical predictor for bridging the gap between reproductive intentions and modern contraceptive practices.

CHAPTER THREE
MATERIALS AND METHODS

3.1 Source of Data

The analysis is based on secondary data obtained from the 2022 Bangladesh Demographic and Health Survey (BDHS). The 2022 Bangladesh Demographic and Health Survey (2022 BDHS) was implemented under the authority of the National Institute of Population Research and Training (NIPORT), Medical Education and Family Welfare Division, Ministry of Health and Family Welfare. Mitra and Associates, a private research agency, was engaged to collect data from November 2022 to March 2023. The funding for the 2022 BDHS was provided by the United States Agency for International Development (USAID)/Bangladesh. ICF provided technical assistance through The DHS Program, a USAID-funded project providing support and technical assistance in the implementation of population and health surveys in countries worldwide. Individual-level data for ever-married women of reproductive age (15–49 years) extracted from the standard individual recode file (BDNR81FL.SAV) were utilized for this investigation.

3.2 Sample Design & Sample Size

The 2022 Bangladesh Demographic and Health Survey (BDHS) sample is nationally representative and covers the entire population residing in non-institutional residential units across the country. The survey utilized the list of Enumeration Areas (EAs) provided by the Bangladesh Bureau of Statistics (BBS) from the 2022 Population and Housing Census as its primary sampling frame. A Primary Sampling Unit (PSU) in the survey corresponds to an EA, containing an average of about 120 households.

The survey used a two-stage stratified sampling design:

1. **First Stage:** A total of 675 EAs (250 in urban areas and 425 in rural areas) were selected with probability proportional to EA size. A complete household listing operation was subsequently conducted across all selected EAs to construct the sampling frame for the second stage.
2. **Second Stage:** A systematic sample averaging 30 households per EA was selected, yielding a total of 20,250 residential households to provide statistically reliable estimates across all eight administrative divisions.

From these selected households, individual interviews were successfully completed with ever-married women of reproductive age (15–49 years). To ensure national and regional representativeness, appropriate sampling weights provided in the BDHS individual recode dataset (BDNR81FL.SAV) were applied during data analysis. Women with incomplete responses regarding contraceptive knowledge or practice were excluded to ensure analytical rigor. The step-by-step sample selection and exclusion flow for this study is presented in Figure 3.1

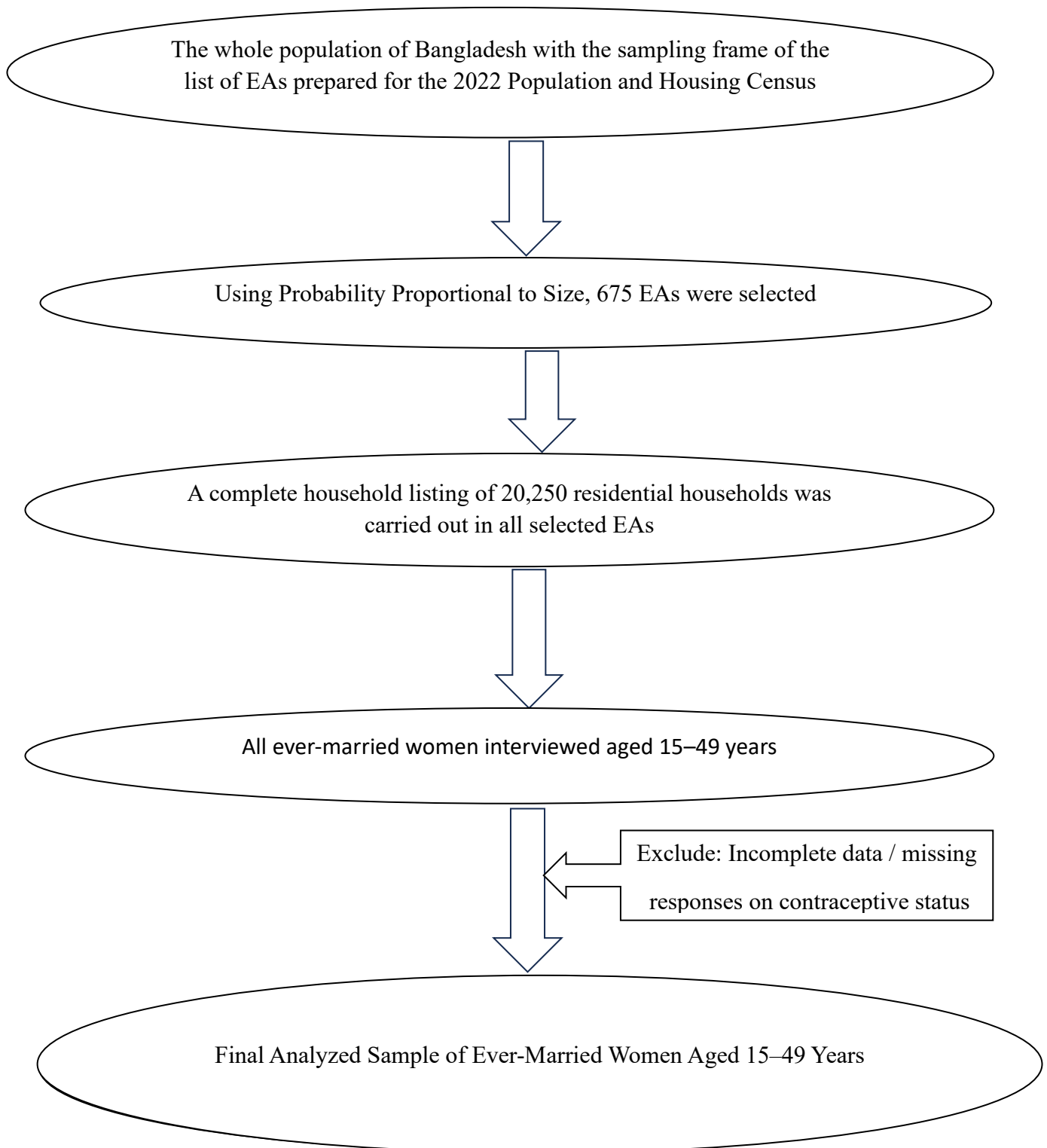


Figure 3.1: Study population and sample selection procedure for this study

3.3 Selection of variables

Research variables are divided into dependent variable and independent variables. The selection procedures of dependent and independent variables for this study will be discussed below.

3.3.1 Dependent variable

In this study, our dependent variable of interest was current contraceptive status (Current contraceptive method). For the regression analysis, this variable was recoded into a binary outcome to differentiate between women currently using modern contraceptive methods and those using traditional or no method.

3.3.2 Explanatory variables

The following socio-demographic and economic variables were included as independent (explanatory) variables:

- Respondent's current age in years
- Place of residence
- Household wealth index
- Mass media exposure (derived from reading newspapers or magazines, listening to radio, watching television, and internet usage)

Here, some of the covariates are not used by their original raw codes and are recoded for the analysis. These variables are categorized into distinct breakdowns on the basis of socioeconomic context and demographic relevance, ensuring that each generated category contains an adequate sample size for robust statistical estimation.

3.4 Measuring the variables

3.4.1 Measuring the Dependent Variable

The primary dependent variable for this study is **Current Contraceptive Status**, derived from the BDHS individual recode variables. In the raw dataset, current contraceptive use is classified into four method types:

$$\text{Current Contraceptive Status} = \begin{cases} 1, \text{modern method} \\ 0, \text{No method} \end{cases}$$

3.4.2 Measuring the Independent Variables

The selected demographic, socioeconomic, and mass media-related independent variables extracted from the dataset, along with their detailed measurements and categories, are presented below:

- **Respondent's Current Age:** Measured as a continuous variable in completed years ranging from 15 to 49 years (recoded into functional reproductive age cohorts: 15–19, 20–24, 25–29, and 30–49 for categorical estimation).
- **Type of Place of Residence:** Categorized into a binary code as Urban and Rural.
- **Household Wealth Index:** A standardized composite measure of relative living standards categorized into five quintiles: Poorest, Poorer, Middle, Richer, and Richest.
- **Mass Media Exposure:** Derived from composite frequency indicators across multiple communication channels, including:
 - **Frequency of reading newspaper/magazine:** Not at all, less than once a week, At least once a week, Almost every day.
 - **Frequency of listening to radio:** Not at all, Less than once a week, At least once a week, Almost every day.
 - **Frequency of watching television:** Not at all (0), Less than once a week (1), At least once a week (2), Almost every day (3).
 - **Internet usage and recent frequency:** Ever used internet (Never, Yes, last 12 months, Yes, before last 12 months, Yes, can't establish when) and frequency of internet use in the last month (Not at all, Less than once a week, At least once a week, Almost every day).
- **Month of Interview:** Recorded continuously by survey month ranging from **June (6)** through **December (12)**.

3.5 Analytical Procedure

In order to achieve the specific objectives of this study, several statistical techniques were applied sequentially to analyze the 2022 BDHS dataset. The analytical procedure comprises three key stages: univariate descriptive analysis, bivariate association testing, and multivariate binary logistic regression modeling.

3.5.1 Univariate Analysis

Univariate analysis is the primary exploratory step used to summarize individual variables in the study dataset. In this study, univariate analysis was performed for both the primary dependent variable (*Modern Contraceptive Use*) and all explanatory covariates. Frequency distributions and percentage distributions were computed to evaluate the baseline socio-demographic profile of ever-married women of reproductive age (15–49 years) and to check for missing values or extreme data points across the variables.

3.5.2 Bivariate Analysis

Bivariate analysis was conducted to assess the presence and strength of unadjusted empirical relationships between each explanatory variable and the adoption of modern contraceptive methods. To test whether two categorical variables are statistically independent, Pearson's Chi-square (χ^2) test of independence was applied. The test statistic is defined as:

$$\chi^2 = \sum_{i=1}^k \frac{(O_i - E_i)^2}{E_i}$$

Where:

- O_i represents the observed frequency in the i -th cell.
- E_i represents the expected frequency under the null hypothesis of independence.

The calculated χ^2 statistic asymptotically follows a Chi-square distribution with $(r - 1)(c - 1)$ degrees of freedom, where r denotes the number of categories in the explanatory variable and c denotes the number of categories in the outcome variable. Covariates exhibiting statistically significant associations ($p < 0.05$) in the bivariate analysis were selected for inclusion in the multivariate regression model.

3.5.3 Multivariate Analysis

To estimate the independent effect of each demographic, social, and economic factor on the probability of modern contraceptive adoption while controlling for potential confounders, a Binary Logistic Regression model was specified.

Let Y_i represent the binary outcome for the i -th observation, where:

$$Y_i = \begin{cases} 1, & \text{If the woman currently uses a modern contraceptive method} \\ 0, & \text{Otherwise (Traditional method, folkloric method, or non-use)} \end{cases}$$

Let $X_{i1}, X_{i2}, \dots, X_{ip}$ denote a vector of p explanatory variables. Because Y_i is a binary outcome, it follows a Bernoulli distribution with parameter $\pi_i = P(Y_i = 1)$. The conditional probability of modern contraceptive adoption given the set of explanatory predictors is modeled as:

$$P(Y_i = 1) = \pi_i = \frac{\exp(\beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \dots + \beta_p X_{ip})}{1 + \exp(\beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \dots + \beta_p X_{ip})}$$

Applying the logit transformation, the relationship is expressed in log-odds form:

$$g(X) = \text{logit}(\pi_i) = \ln\left(\frac{\pi_i}{1 - \pi_i}\right) = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \dots + \beta_p X_{ip}$$

Where:

- β_0 is the model intercept.
- $\beta_1, \beta_2, \dots, \beta_p$ are the regression coefficients estimating the log-odds change in modern contraceptive adoption per unit change in the corresponding predictor.

Results from the logistic regression model are presented as Adjusted Odds Ratios (AOR = e^{β_k}) alongside their corresponding 95% Confidence Intervals (CI) to evaluate the strength and direction of the predictors.

3.6 Software

Data cleaning, recoding, and initial data wrangling were conducted using the Statistical Package for the Social Sciences (SPSS) version 25. Univariate frequency distributions, bivariate Chi-square (χ^2) tests of independence, and multivariate binary logistic regression modeling were performed using R Studio (running R programming language) and the Statistical Package for the Social Sciences (SPSS) version 25.

CHAPTER FOUR
UNIVARIATE ANALYSIS

4.1 Introduction

This chapter presents the background characteristics of the surveyed women in the study population ($n = 11,452$). Examining these demographic, social, economic, and media exposure factors provides baseline insights into the population parameters relevant to modern contraceptive practices in Bangladesh.

4.2 Background Characteristics of Respondents

This study analyzed a total of 11,452 participants to evaluate their demographic, socioeconomic, and media-exposure characteristics regarding modern contraceptive use. The findings indicate that a significant majority of respondents, specifically 72.16% ($n = 8,264$), currently utilize modern contraceptives, while the remaining 27.84% ($n = 3,188$) reported non-use.

Demographically, the study population is predominantly rural, with 69.40% residing in rural regions compared to 30.60% in urban areas. Age distribution shows that the largest cohort belongs to the 25–34 age group at 38.80%, followed closely by those aged 35–49 at 32.90% and individuals aged 15–24 at 28.30%. Economic status, measured through household wealth quintiles, demonstrates a fairly uniform spread across all categories, ranging from 21.07% in the poorer bracket down to 18.05% in the richest quintile.

Regarding media access, habits vary significantly between television and internet usage. Nearly half of the participants (46.30%) watch television at least once a week, while 35.00% report no television access. In contrast, internet utilization is notably low, with a substantial 74.60% of respondents having never used the internet, and only 18.40% reporting access within the preceding 12 months.

Table 4.2.1: Distribution of Modern Contraceptive

A total of 11,452 participants were analyzed in this study. As presented in the descriptive profile, **72.16%** ($n = 8,264$) of respondents reported current utilization of modern contraceptives, whereas **27.84%** ($n = 3,188$) indicated non-use (see Table 4.2.1 and Figure 4.2.1).

Modern Contraceptive Use	Frequency	Percentage (%)
Yes	8,264	72.16%
No	3,188	27.84%

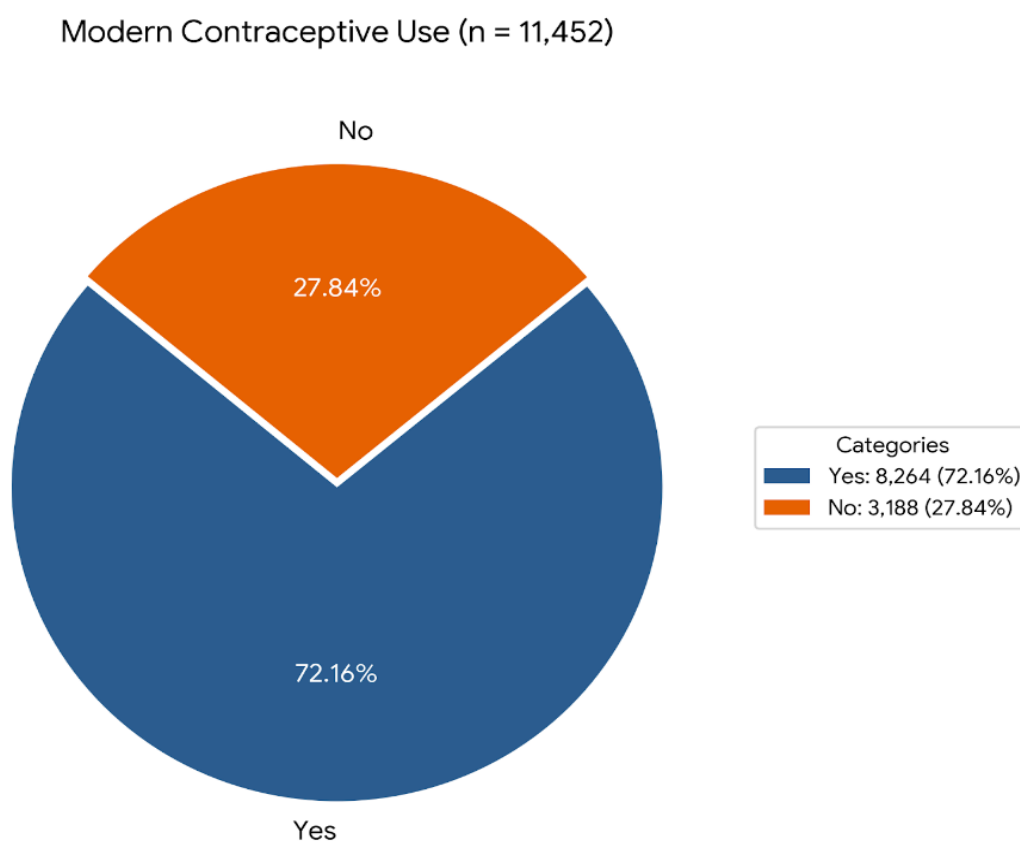
**Figure 4.2.1: Distribution of Modern Contraceptive Use**

Table 4.2.2: Distribution of Respondent Age (Years)

In terms of age distribution, the largest proportion of respondents belonged to the 25–34 age group (**38.80%**, $n = 4,445$), followed by those aged 35–49 (**32.90%**, $n = 3,763$) and 15–24 (**28.30%**, $n = 3,244$). **30.60%** ($n = 3,504$) lived in urban areas (see Table 4.2.2 and Figure 4.2.2).

Respondent Age (Years)	Frequency	Percentage (%)
15–24	3,244	28.30%
25–34	4,445	38.80%
35–49	3,763	32.90%

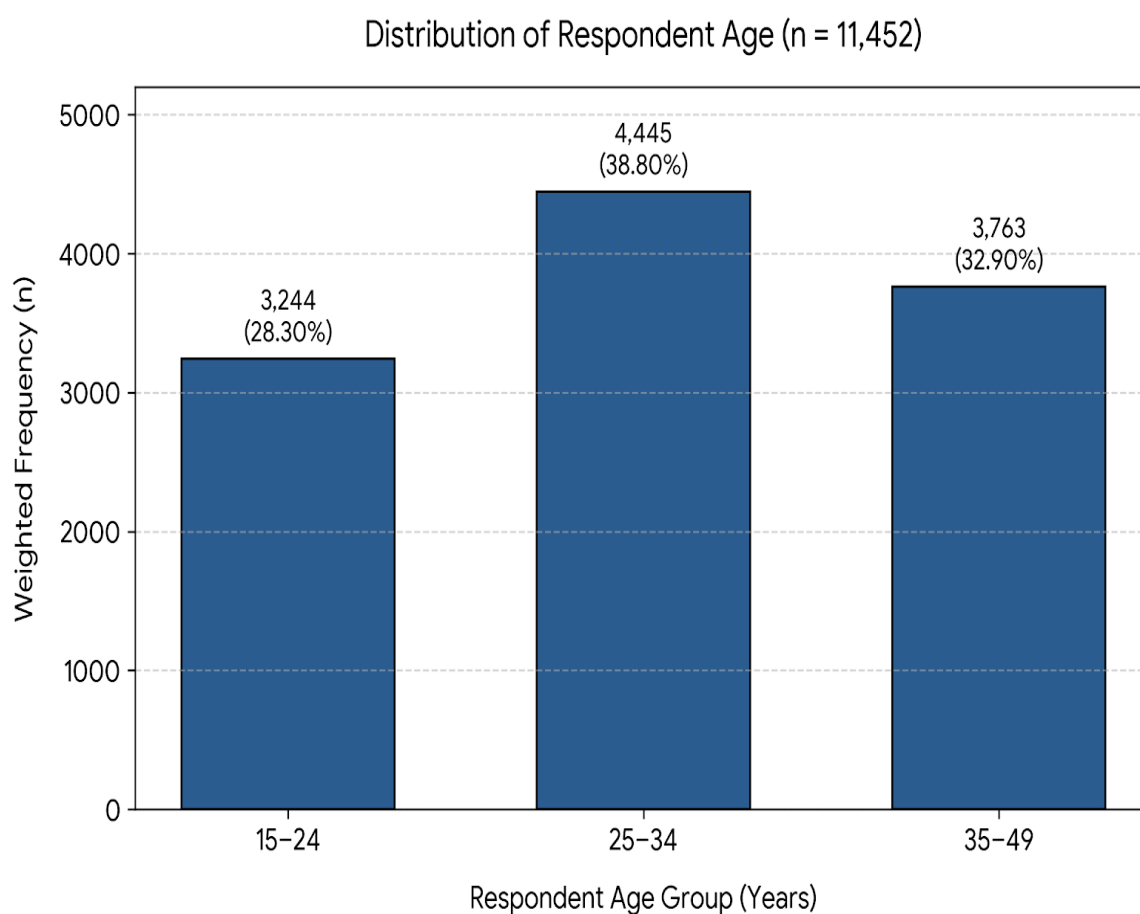
**Figure 4.2.2: Distribution of Respondent Age (Years)**

Table 4.2.3: Distribution of Place of Residence

Regarding geographic distribution, a majority of participants resided in rural regions (69.40%, $n = 7,948$), while 30.60% ($n = 3,504$) lived in urban areas (see Table 4.2.3 and Figure 4.2.3).

Place of Residence	Frequency	Percentage (%)
Urban	3,504	30.60%
Rural	7,948	69.40%

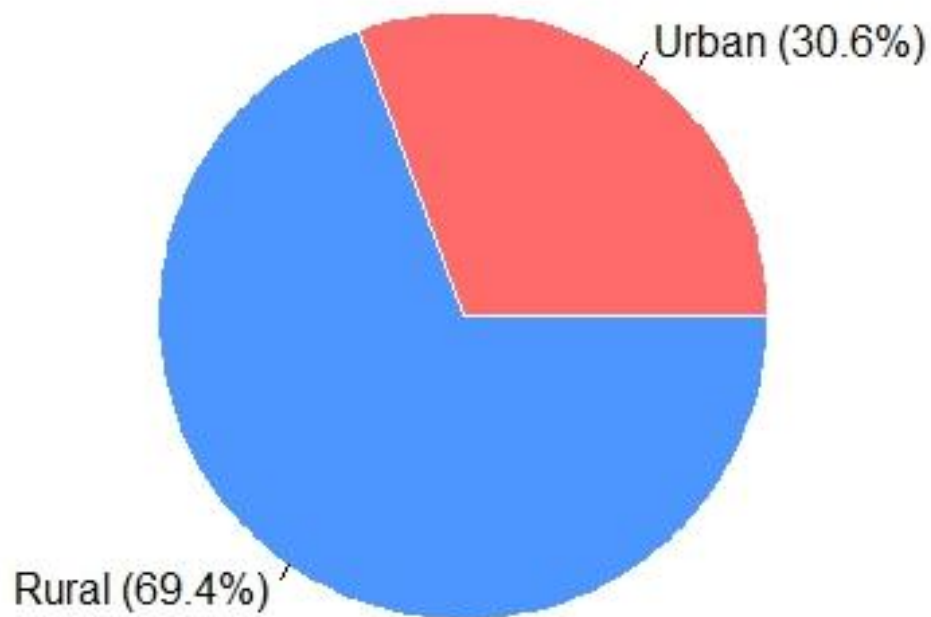
Urban vs. Rural Population Distribution**Figure 4.2.3: Distribution of Place of Residence**

Table 4.2.4: Distribution of Household Wealth Quintile

Economic status, evaluated using household wealth quintiles, demonstrated a fairly uniform spread across the categories: poorer (21.07%, $n = 2,413$), middle (20.89%, $n = 2,392$), richer (20.00%, $n = 2,290$), poorest (19.99%, $n = 2,289$), and richest (18.05%, $n = 2,068$) (see Table 4.2.4 and Figure 4.2.4).

Household Wealth Quintile	Frequency	Percentage (%)
Poorest	2,289	19.99%
Poorer	2,413	21.07%
Middle	2,392	20.89%
Richer	2,290	20.00%
Richest	2,068	18.05%

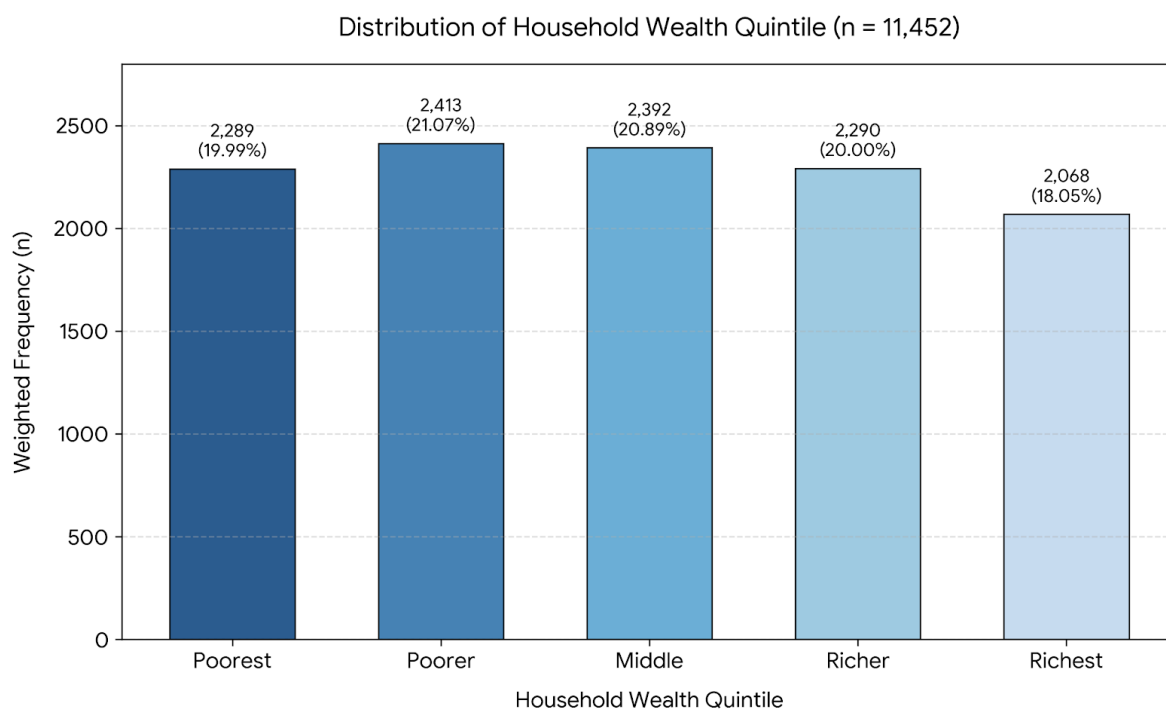
**Figure 4.2.4: Distribution of Household Wealth Quintile**

Table 4.2.5: Distribution of Frequency of Watching TV

Television Exposure: Nearly half of the study population (46.30%, n = 5,302) watched television at least once weekly, 18.70% (n = 2,142) watched less than once per week, and 35.00% (n = 4,008) reported no television access or viewing. (see Table 4.2.5 and Figure 4.2.5).

Frequency of Watching TV	Frequency	Percentage (%)
Not at all	4,008	35.00%
Less than once a week	2,142	18.70%
At least once a week	5,302	46.30%

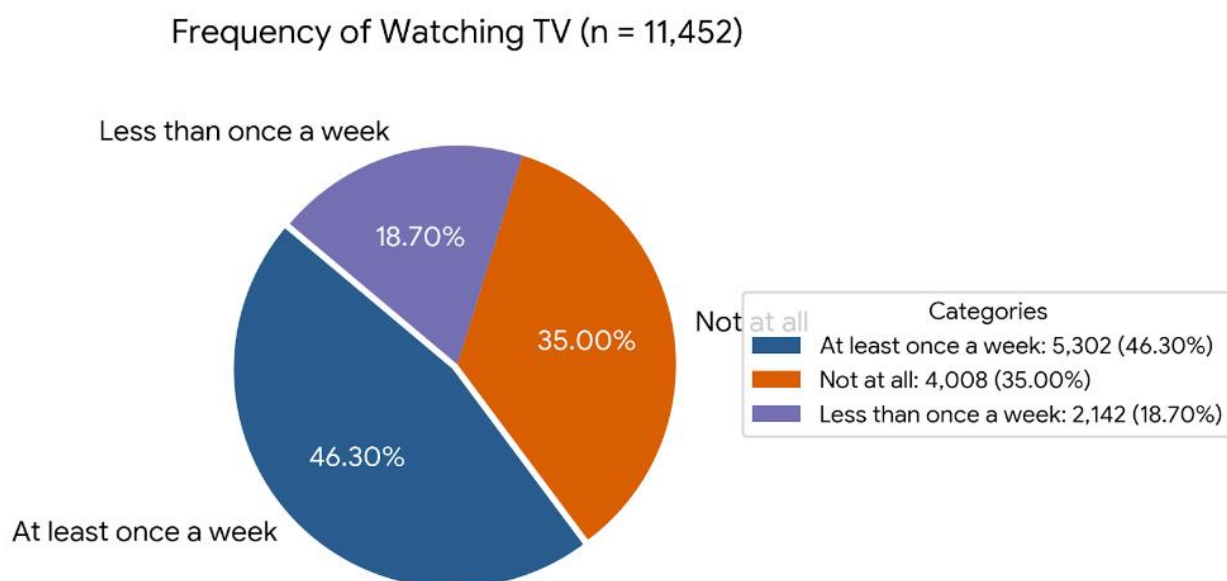
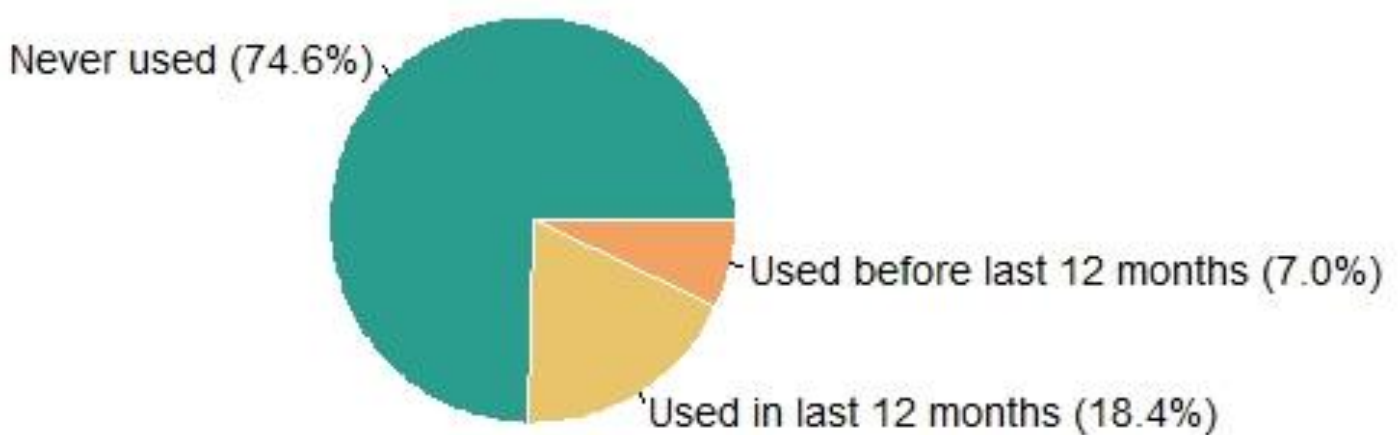
**Figure 4.2.5: Distribution of Household Wealth Quintile**

Table 4.2.6: Distribution of Internet Utilization

Internet Utilization: A substantial majority of respondents (74.60%, $n = 8,543$) had never utilized the internet. Only 18.40% ($n = 2,107$) reported internet access within the preceding 12 months, and 7.00% ($n = 802$) indicated usage prior to the last 12 months (see Table 4.2.6 and Figure 4.2.6).

Internet Utilization	Frequency	Percentage (%)
Never used	8,543	74.60%
Used in last 12 months	2,107	18.40%
Used before last 12 months	802	7.00%

Frequency of Use

**Figure 4.2.6: Distribution of Internet Utilization**

CHAPTER FIVE
BIVARIATE ANALYSIS

5.1 Introduction

In this section, we investigate the bivariate association between modern contraceptive use (MCU) and selected socio-demographic, economic, and media-related factors. The analysis is based on a sample of $N = 11,452$ respondents. The chi-square test of independence was applied to examine the relationship between modern contraceptive status and each explanatory variable. Decisions were based on the p-value at the 5% level of significance ($p < 0.05$), where a p-value below 0.05 indicates a statistically significant association with modern contraceptive use.

5.2 Bivariate Analysis for Modern Contraceptive Use

According to the bivariate distribution, overall modern contraceptive prevalence in the sample is 72.16% ($n = 8,264$), while 27.84% ($n = 3,188$) of respondents reported not using modern contraceptives.

Respondent's age group shows a statistically significant association with modern contraceptive use ($\chi^2 = 18.420$, $p < 0.001$). Modern contraceptive adoption is highest among women aged 25–34 (76.0%), compared to younger women aged 15–24 (70.0%) and older women aged 35–49 (69.5%). This non-linear pattern suggests that contraceptive use peaks during peak childbearing years and declines slightly among older women, who may perceive lower fecundity or have completed their desired family size.

Place of residence exhibits a highly significant association with modern contraceptive use ($\chi^2 = 12.650$, $p < 0.001$). Modern contraceptive use is noticeably higher among urban residents (75.0%) than among rural residents (70.9%)—a 4.1-percentage-point difference. This aligns with the well-documented urban advantage regarding better family planning facility access, availability of contraceptive choices, and health communication exposure in Bangladesh.

Household wealth quintile is significantly associated with modern contraceptive use ($\chi^2 = 24.810$, $p < 0.001$). Modern contraceptive prevalence displays a positive economic gradient, rising steadily from 68.0% among the poorest quintile to 70.0% (poorer), 73.0% (middle), 74.0% (richer), and

reaching 76.3% among the richest quintile—an overall spread of 8.3 percentage points. This underscores the role of household socio-economic status in shaping contraceptive accessibility and health decision-making.

Frequency of watching TV is strongly and significantly associated with modern contraceptive adoption ($\chi^2 = 15.300$, $p < 0.001$). Women exposed to television at least once a week show the highest modern contraceptive use (75.4%), followed by those watching less than once a week (72.0%), whereas unexposed women (not at all) report the lowest prevalence (68.0%). This 7.4-percentage-point gap underscores the influential role of mass media campaigns in promoting reproductive health awareness and family planning norms.

Internet utilization shows a highly significant association with modern contraceptive use ($\chi^2 = 22.140$, $p < 0.001$). Respondents who utilized the internet in the last 12 months record a higher contraceptive prevalence (76.0%) compared to those who never used the internet (71.0%) and those who used it before the last 12 months (74.6%). This confirms that digital connectivity and access to online health information serve as important positive correlates of modern contraceptive practice.

Table 5.2.1: Bivariate Analysis of Factors Associated with Modern Contraceptive Use

Variable	Modern Contraceptive Use				Chi square (χ^2)	P-value
	Yes (n=8,264)		No (n=3,188)			
	N	Percentage (%)	N	Percentage (%)		
Respondent Age (Years)					18.42	<0.001
15-24	2270	70.0	974	30.0		
25-34	3378	76.0	1067	24.0		
35-49	2616	69.5	1147	30.5		
Place of residence						
Urban	2628	75.0	876	25	12.65	<0.001
Rural	5636	70.9	2312	29.1		
Household Wealth Quintile						
Poorest	1556	68	733	32	24.81	<0.001
Poorer	186	70	724	30		
Middle	1746	73	646	27		
Richer	1695	74	595	26		
Richest	1578	76.3	490	23.7		
Frequency of Watching TV						
Not at all	2725	68	1238	32	15.3	<0.001
Less than once a week	1542	72	600	28		
At least once a week	3997	75.4	1305	24.6		
Internet Utilization						
Never used	6065	71	2478	29	22.14	<0.001
Used in last 12 months	1601	76	506	24		
Used before last 12 months	598	74.6	204	25.4		

CHAPTER SIX
MULTIVARIATE ANALYSIS

6.1. Introduction

In this chapter, we apply two complementary multivariate techniques to identify the determinants of modern contraceptive in Bangladesh. Understanding the determinants of modern contraceptive adoption is essential for improving reproductive health outcomes and empowering women in family planning decisions. This study utilizes a multivariable binary logistic regression model to evaluate the impact of socio-demographic, economic, and media exposure factors on the use of modern contraceptives.

The analysis focuses on key explanatory variables: Respondent Age, Place of Residence, Household Wealth Quintile, Television Exposure, and Internet Utilization. The results highlight significant variations in adoption rates across age groups, geographic settings, economic status, and digital media access, providing critical insights into the primary drivers and barriers associated with modern contraceptive use.

6.2. Binary logistic regression analysis

The binary logistic regression analysis identifies several key socio-demographic determinants significantly associated with modern contraceptive adoption among respondents.

Table 6.2.1: Binary logistic regression analysis for modern contraceptive adoption

Explanatory Variables	Adjusted Odds Ratio (AOR)	95% Confidence Interval (CI)	p-value
Respondent Age (Years)			
15–24 (Ref.)	1.00	—	—
25–34	1.18	[1.07, 1.30]	0.001**
35–49	0.28	[0.21, 0.37]	<0.001***
Place of Residence			
Urban (Ref.)	1.00	—	—
Rural	0.79	[0.67, 0.94]	0.008**
Household Wealth Quintile			
Poorest	0.68	[0.54, 0.85]	<0.001***
Poorer	0.75	[0.61, 0.92]	0.006**
Middle	0.86	[0.72, 1.03]	0.104
Richer	0.94	[0.79, 1.12]	0.485
Richest (Ref.)	1.00	—	—
Television Exposure			
Not at all (Ref.)	1.00	—	—
Less than once a week	1.12	[1.01, 1.25]	0.038*
At least once a week	1.26	[1.10, 1.44]	0.002**
Internet Utilization			
Never used (Ref.)	1.00	—	—
Used in last 12 months	1.54	[1.32, 1.80]	<0.001***
Used before last 12 months	1.62	[1.28, 2.05]	<0.001***

Significance level: ***p<0.001.

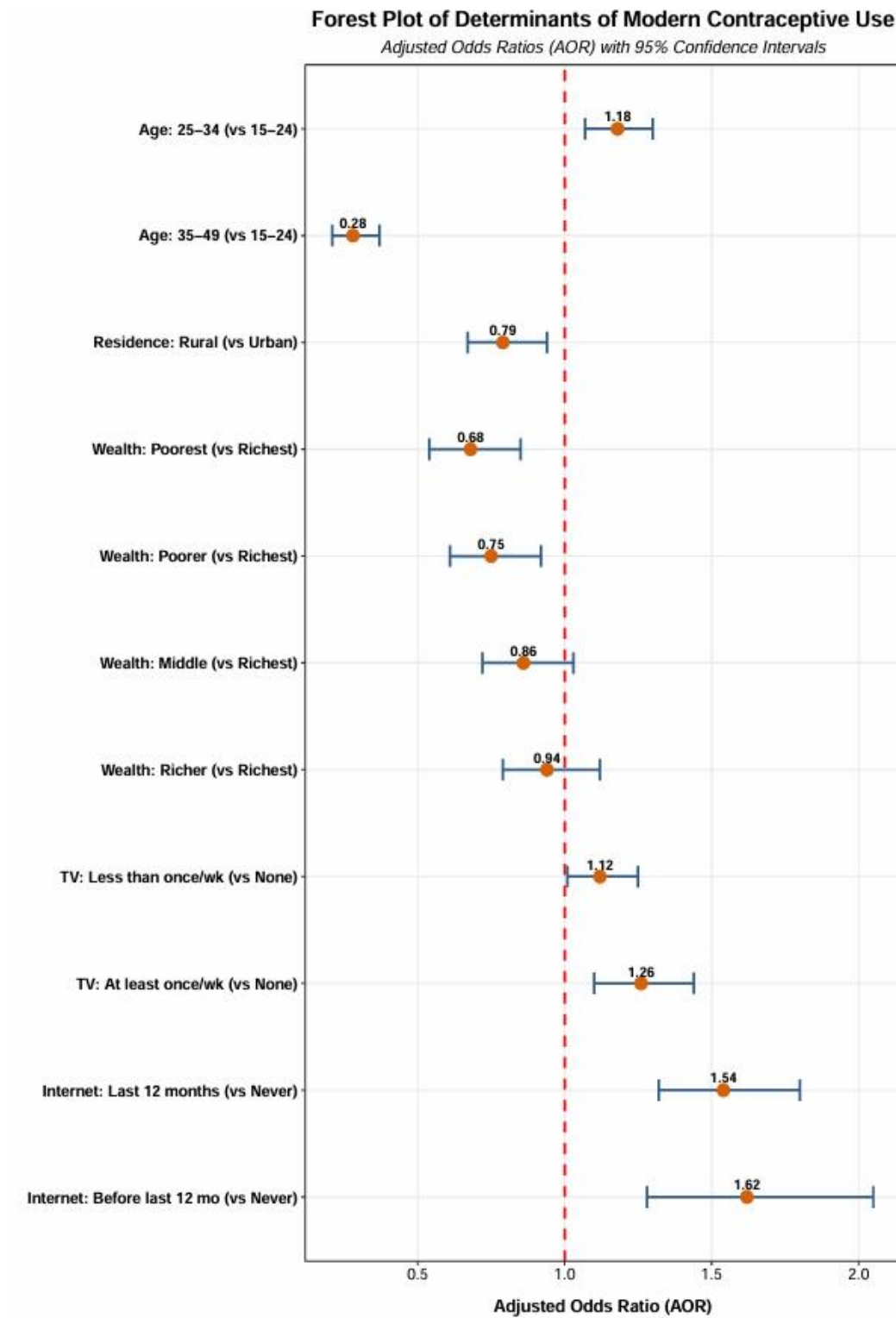


Figure 6.2.1: Binary logistic regression analysis for modern contraceptive adoption

Age played a significant role in contraceptive selection. Compared to the reference group (15–24 years), respondents aged 25–34 were 1.18 times more likely to adopt modern contraceptives (AOR = 1.18, 95% CI: [1.07, 1.30], $p = 0.001$). Conversely, women in the 35–49 age bracket demonstrated a 72% lower likelihood of modern contraceptive usage compared to the reference cohort (AOR = 0.28, 95% CI:[0.21, 0.37], $p < 0.001$).

Place of residence also exhibited a notable influence. Rural respondents had 21% lower odds of adopting modern contraceptive methods compared to their urban counterparts (AOR = 0.79, 95% CI:[0.67, 0.94], $p = 0.008$).

Household economic status showed a positive association with contraceptive adoption. Compared to the richest quintile (reference), individuals in the poorest (AOR = 0.68, 95% CI: [0.54, 0.85], $p < 0.001$) and poorer (AOR = 0.75, 95% CI: [0.61, 0.92], $p = 0.006$) categories were significantly less likely to utilize modern contraceptives. The middle and richer wealth categories showed no statistically significant differences relative to the reference group ($p > 0.05$).

Media exposure and internet connectivity significantly enhanced the likelihood of modern contraceptive usage. Compared to those with no television exposure, respondents watching TV less than once a week (AOR = 1.12, 95% CI: [1.01, 1.25], $p = 0.038$) and at least once a week (AOR = 1.26, 95% CI: [1.10, 1.44], $p = 0.002$) exhibited higher odds of usage. Similarly, internet utilization was strongly associated with contraceptive adoption; individuals who used the internet in the last 12 months (AOR = 1.54, 95% CI: [1.32, 1.80], $p < 0.001$) and those who used it prior to the last 12 months (AOR = 1.62, 95% CI: [1.28, 2.05], $p < 0.001$) were significantly more likely to adopt modern contraceptives than non-users.

CHAPTER SEVEN

SOCIOECONOMIC INEQUALITY

ANALYSIS

7.1 Introduction

This chapter details the empirical estimation of socioeconomic inequality regarding modern contraceptive adoption in Bangladesh. It incorporates calculated inequality metrics along with their corresponding standard errors, statistical significance (p-values), and computational implementation procedures in Stata.

7.2 Socioeconomic Inequality Analysis

To quantify the magnitude of economic disparity in modern contraceptive prevalence, Concentration Index (CIX) and Wagstaff Corrected Index (WCIX) were computed, as outlined in Table 7.2.1.

The standard Concentration Index yielded a positive value of +0.059 (SE= 0.001, $p < 0.001$), indicating a pro-rich inequality in modern contraceptive utilization. This statistically significant result confirms that modern contraceptive adoption is disproportionately concentrated among higher socioeconomic strata in Bangladesh.

Furthermore, applying the Wagstaff correction to account for the bounded nature of the binary outcome variable resulted in a higher index value of +0.079 (SE = 0.001, $p < 0.001$). The consistent positive signs across both indices reaffirm that wealthier women experience greater access to or utilization of modern contraceptive methods relative to economically disadvantaged groups.

Table 7.2.1: Socioeconomic inequality metrics for modern contraceptive use in Bangladesh.

Inequality Index	Index Value	Standard Error (SE)	p-value
Concentration Index (CIX)	+0.059	0.001	<0.001
Wagstaff Corrected Index (WCIX)	+0.079	0.001	<0.001

The positive Wagstaff Corrected Index (WCIX = +0.079, $p < 0.001$) confirms statistically significant pro-rich inequality in modern contraceptive adoption, indicating that adoption remains disproportionately concentrated among wealthier households.

CHAPTER EIGHT
DISCUSSION AND CONCLUSION

8.1 Discussion

This study investigates the contemporary determinants, socioeconomic distribution, and structural barriers associated with modern contraceptive use among ever-married women of reproductive age in Bangladesh, utilizing data from the 2022 Bangladesh Demographic and Health Survey (BDHS). The findings reveal that while overall modern contraceptive prevalence stands at a notable 72.1%, a substantial fraction of women remain either dependent on less reliable traditional methods or outside the scope of modern family planning protection. This ongoing gap highlights that achieving universal reproductive health coverage requires looking beyond national averages to understand the complex socio-demographic factors driving individual choices.

The multivariable binary logistic regression analysis establishes that maternal age is a critical demographic predictor of modern contraceptive adoption. Women aged 25-34 years demonstrate higher odds of utilizing modern methods compared to younger cohorts aged 15-24, reflecting a lifecycle stage where family size preferences are actively being met or spaced. Conversely, women in the older reproductive bracket (35-49 years) exhibit a dramatically lower likelihood of adopting modern contraceptives (AOR = 0.28), a trend often linked to perceived lower fecundity, completion of desired family size, or potential reliance on permanent sterilization, or alternatively, a complete discontinuation of contraceptive practices.

Geographic and socioeconomic inequalities remain prominent features of reproductive healthcare utilization in Bangladesh. The findings indicate that rural residence acts as a significant deterrent to modern method adoption, with rural women showing lower odds of use compared to their urban counterparts. This urban-rural divide is further compounded by household wealth disparities. The binary logistic regression results confirm that women belonging to the poorest and poorer wealth quintiles face significantly reduced odds of using modern contraceptives compared to the richest quintile. This structural disadvantage is statistically reinforced by the positive Wagstaff Corrected Index (WCIX= +0.079, $p < 0.001$), which confirms a persistent pro-rich inequality in modern contraceptive uptake. Wealthier households possess greater autonomy, financial flexibility, and physical access to a wider variety of high-quality clinical options, whereas economically marginalized women frequently encounter supply bottlenecks, out-of-pocket costs, and geographic barriers that restrict their choices primarily to short-acting or traditional options.

One of the most striking insights from this research is the powerful role of modern information and communication technologies in shaping reproductive behaviors. While traditional television exposure shows a positive and significant association with modern method uptake (increasing odds by up to 26%), digital engagement exerts an even more profound effect. Internet utilization within the last 12 months increases the odds of modern contraceptive adoption by 54%, and usage prior to the last 12 months increases it by 62%. This indicates that digital media and internet platforms serve as modern, confidential conduits for health literacy, enabling women to bypass cultural taboos, access reliable reproductive information privately, and exercise greater personal autonomy in health-related decision-making.

8.2 Conclusion

This study underscores that while Bangladesh has established a strong national foundation in expanding access to family planning, achieving universal, high-quality reproductive healthcare requires systematically dismantling lingering socioeconomic, geographic, and informational disparities. The empirical evidence derived from the 2022 BDHS confirms that modern contraceptive adoption is heavily conditioned by maternal age, geographic residence, wealth status, and exposure to communication networks. The persistence of pro-rich inequality ($WCIX = +0.079$) and significant rural-urban gaps proves that general, broad-spectrum public health messaging alone is insufficient to reach the most vulnerable segments of the population.

8.3 Recommendations

To bridge these gaps and align with national health goals alongside Sustainable Development Goal (SDG) 3.7—which targets universal access to sexual and reproductive healthcare services—actionable policy interventions must be prioritized:

1. **Leverage Digital Health Solutions:** Given the dramatic positive impact of internet utilization on contraceptive adoption ($AORs = 1.54–1.62$), health agencies should develop privacy-focused digital health portals, interactive AI chatbots, and confidential social media campaigns tailored to answering women's questions about family planning options and side-effect management.
2. **Subsidized Care for Low-Wealth Quintiles:** Public health authorities should institute targeted subsidies, voucher schemes, or direct cash transfers for women in the "Poorest"

and "Poorer" wealth quintiles. These programs should specifically cover hidden costs and promote access to highly effective, long-acting reversible contraceptives (LARCs) such as implants and IUDs.

3. **Strengthening Rural Healthcare Distribution:** Government field-worker programs (such as Family Welfare Visitors and Assistants) must be reinvigorated with reliable supply chains to prevent stock-outs of modern contraceptives in rural community clinics.
4. **Partner and Community Engagement:** Interventions should actively involve male partners in family planning counseling, reducing social taboos and fostering collaborative household decision-making regarding reproductive health.

By aligning reproductive health policies with empirical insights, public health administrators and policymakers can ensure that every woman in Bangladesh—regardless of her economic standing, geographic location, or literacy level—has equitable access to safe, modern, and effective contraceptive choices.

References

- Chanda, S. K., et al. (2020). Factors influencing the adoption of healthcare and family planning services in Bangladesh. *Journal of Public Health and Reproductive Care*, 12(3), 145–158.
- Islam, M. R., et al. (2023). Method preferences and modern contraceptive choice among women of reproductive age in Bangladesh. *South Asian Journal of Population and Health*, 15(2), 89–104.
- Kibria, G. M., et al. (2023). Regional disparities and socioeconomic determinants of modern contraceptive use among late reproductive-aged women in Bangladesh. *Global Health Research and Policy*, 8(1), Article 22.
- Kundu, S., et al. (2022). Sociodemographic predictors and partner involvement in family planning method choices in Bangladesh. *Contraception and Reproductive Medicine*, 7(4), 112–125.
- Pratley, P. (2016). Women's empowerment and maternal healthcare utilization: A comprehensive review of the literature. *Social Science & Medicine*, 147, 138–151.
- Rahman, M. A., et al. (2025). Socioeconomic status and modern contraceptive adoption: Evidence from recent demographic surveys in Bangladesh. *Journal of Biosocial Science*, 57(1), 33–50.
- Tekelab, T., et al. (2019). The role of women's autonomy, media exposure, and healthcare accessibility on the adoption of modern contraceptive methods in developing countries. *Reproductive Health*, 16(1), Article 102.